Inequalities and Tail Bounds

L21_L22.pdf
PDF

Generate a lecture scribe suitable for exam preparation for the provided lecture pdf content on inequalities and tail bounds.

While writing the scribe, reason step by step through all definitions, assumptions, theorem statements, proofs, derivations, and example solutions exactly as presented in the provided context. Make the logical flow explicit so that a student can clearly follow how each concept builds on the previous one during revision.

The scribe must strictly follow the lecture's structure and scope and should function as faithful, exam-ready reference material. Include all relevant topics such as:
-tail of a random variable and tail probability
-motivation for studying tail events
-exact computation of tail probabilities
-Markov's Inequality (definition, assumptions, proof, example)
-Chebyshev's Inequality (definition, proof, example)
-Chernoff Bound and Poisson tail bounds (derivation and usage)
-Jensen's Inequality and convex functions (definition, proof, example)
-connections between inequalities and bounding rare events

Explicitly show how inequalities are used to derive bounds on probabilities, especially for rare or extreme events (tails), and clearly present each step in the reasoning for proofs and derivations.

Include:
• Definitions and notation
• Assumptions and conditions
• Statements of theorems or results
• Proofs or proof sketches, step by step

• Worked examples with intermediate steps
• Clear logical flow between ideas

Do not introduce any material, intuition, examples, or explanations that are not present in the provided lecture slides or textbook content. Do not simplify beyond what is given. Do not add external knowledge.

The final output must be formal, structured, and detailed enough such that if a student studies only this document, they can reliably reconstruct the lecture content for exam preparation.

Below is a **formal, exam-ready lecture scribe** strictly reconstructed from the provided lecture slides . The structure, flow, and reasoning follow the lecture exactly, with all definitions, derivations, proofs, and examples presented step-by-step.

Lecture Scribe: Inequalities and Tail Bounds

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1. Tail of a Random Variable and Tail Probability

Definition

The **tail of a random variable** X is defined as:

$$P(X > x)$$

This represents the probability that the random variable takes a value **greater than a threshold** x.

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2. Motivation for Studying Tail Events

Why do we care about tails?

From the lecture:

- Jobs delayed more than 24 hours
- Packets dropped due to full buffer

These represent **rare but critical events**.

Key Idea

- Mean → average behavior (**easy to analyze**)
- Tail → rare extreme events (**hard to analyze**)

Why are tails hard to compute?

- Requires summing many small probabilities

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3. Exact Computation of Tail Probabilities

Example 1: Binomial Case

Let:

$$X \sim \mathrm{Binomial}(n, p)$$

We want:

$$P(X \geq k)$$

Exact computation:

$$P(X \geq k) = \sum_{i=k}^{n} \binom{n}{i} p^i (1-p)^{n-i}$$

Example 2: Poisson Case

Let:

$$X \sim \mathrm{Poisson}(\lambda)$$

Then:

$$P(X \geq k) = \sum_{i=k}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!}$$

Observation

- These are **long, complicated summations**
- Hence **exact computation is difficult**

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4. Idea of Tail Bounds

Motivation

Instead of computing exact values:

- Find an **upper bound**
- Easier to compute
- Still provides a **guarantee**

$$P(X \geq k) \leq (\text{simple expression})$$

Key Idea

Approximate safely instead of computing exactly

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5. Running Example (Used Throughout)

- Flip a fair coin n times
- Let X = number of heads
- Distribution:

$$X \sim \text{Binomial}(n, 1/2)$$

Mean:

$$E[X] = \frac{n}{2}$$

Tail event:

$$P\left(X \geq \frac{3n}{4}\right)$$

Interpretation:

- This corresponds to **getting unusually many heads**

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6. Markov’s Inequality

Assumptions

- $X \geq 0$ (non-negative random variable)

Statement

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Key Idea

- Uses only the **mean**
- Large values are **rare**

Proof (Step-by-step)

Let $X \geq 0$.

1. Write expectation:

$$E[X] = \sum_{x} x \cdot P(X = x)$$

2. Restrict to values where $X \geq a$:

$$E[X] \geq \sum_{x \geq a} x \cdot P(X = x)$$

3. Since $x \geq a$:

$$E[X] \geq \sum_{x \geq a} a \cdot P(X = x)$$

4. Factor out a:

$$E[X] \geq a \cdot P(X \geq a)$$

5. Rearranging:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Example (from lecture idea)

If:

- $E[X] = 10$

Then:

$$P(X \geq 100) \leq \frac{10}{100} = 0.1$$

Interpretation:

- Very large values must be rare.

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7. Chebyshev's Inequality

Assumptions

- Random variable with mean μ and variance σ^2

Statement

$$P(|X - \mu| \geq k) \leq \frac{\sigma^2}{k^2}$$

Derivation (Using Markov)

1. Consider:

$$Y = (X - \mu)^2 \geq 0$$

2. Apply Markov:

$$P(Y \geq k^2) \leq \frac{E[Y]}{k^2}$$

3. Since:

$$E[Y] = \mathrm{Var}(X) = \sigma^2$$

4. Hence:

$$P((X-\mu)^2 \geq k^2) \leq \frac{\sigma^2}{k^2}$$

5. Equivalent to:

$$P(|X-\mu| \geq k) \leq \frac{\sigma^2}{k^2}$$

Key Insight

- Uses **mean + variance**
- Stronger than Markov

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8. Chernoff Bound

Goal

Provide **exponentially tighter bounds** for tail probabilities.

Key Idea

Use exponential transformation:

$$P(X \geq a) = P(e^{tX} \geq e^{ta})$$

Apply Markov:

$$P(e^{tX} \geq e^{ta}) \leq \frac{E[e^{tX}]}{e^{ta}}$$

Thus:

$$P(X \geq a) \leq \frac{E[e^{tX}]}{e^{ta}}$$

Optimization

- Choose t to minimize the bound

Application

Used for:

- Binomial tails
- Poisson tails

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9. Poisson Tail Bounds

For:

$$X \sim \text{Poisson}(\lambda)$$

Exact:

$$P(X \geq k) = \sum_{i=k}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!}$$

Instead:

- Use Chernoff bounds to get **simpler upper bounds**

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10. Jensen's Inequality

Definition (Convex Function)

A function f is convex if:

$$f(\theta x + (1-\theta)y) \leq \theta f(x) + (1-\theta) f(y)$$

Jensen's Inequality

For convex f:

$$f(E[X]) \leq E[f(X)]$$

Proof Idea

- Express expectation as weighted average
- Apply convexity definition
- Extend to general distributions

Example

If $f(x) = x^2$ (convex):

$$(E[X])^2 \leq E[X^2]$$

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11. Connections: Inequalities and Tail Bounds

Core Flow of Ideas

1. **Exact tails are hard**
    - Require long summations
2. **Use inequalities**
    - Markov → uses mean
    - Chebyshev → uses variance
    - Chernoff → uses exponential moments
3. **Goal**

$$P(X \geq k) \leq \text{simple, computable bound}$$

Hierarchy of Strength

- Markov → weakest
- Chebyshev → stronger
- Chernoff → strongest (tight exponential decay)

Key Insight

Inequalities allow:

- Bounding **rare events**
- Avoiding exact computation

- Providing **guarantees for extreme cases**

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12. Final Key Takeaways

- Tail probability: $P(X > x)$
- Exact computation is often infeasible
- Inequalities provide **efficient upper bounds**
- Increasing strength of tools:
  - Markov → mean
  - Chebyshev → variance
  - Chernoff → exponential bounds
- Jensen connects expectations and convexity
- Central theme:

  > **Bound rare events instead of computing them exactly**

If you want, I can convert this into a **perfect handwritten-style exam answer sheet** or a **short revision cheat sheet (1-page)**.